SMART INDIA HACKATHON 2026 | INDIAN AVIATION GOVERNANCE & ECONOMETRIC STANDARDS

Airfare Price Index (APIx): Research & Methodology

Statistical Formulation, Indian Regulatory Grounding, Weighting Mechanics, and International Precedents

TARGET INDEX	MATHEMATICAL MODEL	WEIGHTING SOURCE	BENCHMARK BASE
APIx (Domestic Airfare)	Modified Laspeyres / Lowe	DGCA Quarterly PAX Filings	Sept 2022 = 100.00

1. STATISTICAL FOUNDATION

1.1 Textbook Laspeyres Price Index

In neoclassical price index theory, a standard **Laspeyres Price Index** measures the relative percentage change in the cost of purchasing a fixed basket of goods and services evaluated at base-period consumption quantities (Laspeyres, 1871):

STANDARD LASPEYRES FORMULA (TEXTBOOK FORM)

$$I_L = \frac{\Sigma_{i=1}^{n} P(i, t) \times Q(i, 0)}{\Sigma_{i=1}^{n} P(i, 0) \times Q(i, 0)} \times 100$$

Where **P(i, t)** is the price of item *i* in period *t*; **P(i, 0)** is the price in base period 0; and **Q(i, 0)** is the fixed consumption volume during base period 0.

1.2 The APIx Formulation

In civil aviation, an "item" represents a sovereign origin–destination city pair sampled across distinct advance booking horizons $h \in \{T+1, T+7, T+15, T+30, T+45\}$. APIx implements a **Modified Laspeyres Price Index** weighted by passenger volume shares derived from Directorate General of Civil Aviation (DGCA) quarterly traffic filings:

APIX MODIFIED LASPEYRES FORMULATION

$$APIx_t = \left[\Sigma_{r=1}^{R} w_{r,0} \times \frac{P(r, t)}{P(r, 0)} \right] \times 100$$

$$\text{where passenger weight:} \quad w_{r,0} = \frac{PAX(r, 0)}{\Sigma_{k=1}^{R} PAX(k, 0)} \quad \text{such that} \quad \Sigma_{r=1}^{R} w_{r,0} = 1.000000$$

Where **P(r, t)** is the IQR-cleaned median fare on route *r* at date *t*; **P(r, 0)** is the baseline reference median fare (Sept 2022 = 100.0); and **PAX(r, 0)** is the official quarterly passenger count recorded in DGCA filings.

1.3 Methodological Equivalence

DIMENSION	TEXTBOOK LASPEYRES	U.S. BLS AIR PASSENGER FARES INDEX	APIX (INDIA DOMESTIC INDEX)
Aggregator	Fixed-base arithmetic mean	Modified Laspeyres (Passenger-Revenue Weighted)	Modified Laspeyres (Passenger-Volume Weighted)
Weighting Source	Base quantities Q_0	DOT DB1B Survey & Commerce Dept I-92	DGCA Form A/B Quarterly Traffic Returns
Formula Structure	$[\Sigma P_t Q_0 / \Sigma P_0 Q_0] \times 100$	$\Sigma w_{r,0} \cdot [P(r,t) / P(r,0)] \times 100$	$\Sigma w_{r,0} \cdot [P(r,t) / P(r,0)] \times 100$
Precedent Reference	Classical Economics (1871)	U.S. BLS IPP Handbook (Ref #6)	Directly Method-Equivalent to BLS IPP (Ref #6)

1.4 Operational Reality: The Lowe Index Distinction

As documented in the **UK Office for National Statistics (ONS) Consumer Price Indices Technical Manual** (Reference #9), when quantity weights Q_b originate from an independent base period $b \neq 0$, the index is formally classified as a **Lowe Index**:

LOWE PRICE INDEX FORMULATION (UK ONS SPECIFICATION)

$$I_{Lowe,\ t} = \frac{\Sigma_{r=1}^{R}\ P(r,\ t) \times Q(r,\ b)}{\Sigma_{r=1}^{R}\ P(r,\ 0) \times Q(r,\ b)} = \Sigma_{r=1}^{R}\ w_{r,b} \times \frac{P(r,\ t)}{P(r,\ 0)}$$

Because DGCA passenger volume matrices are refreshed quarterly with a minor reporting lag, APIx operates in practical execution as a **Lowe Index**, matching international best practices across the UK ONS, Eurostat, and the U.S. BLS.

2. WHY WEIGHTING MATTERS: STATISTICAL JUSTIFICATION

2.1 Variance & Outlier Sensitivity: Weighted vs. Unweighted Indices

An unweighted arithmetic mean assigns equal weight (*1/R*) to every route regardless of passenger volume:

NAIVE UNWEIGHTED MEAN INDEX

$$Index_{Naive,\ t} = \frac{1}{R}\ \Sigma_{r=1}^{R}\ \frac{P(r,\ t)}{P(r,\ 0)} \times 100$$

Theoretical Proof of Distortion (80-Route Network):

Suppose 79 trunk routes experience zero price movement (P_t/P_0 = 1.00), while one low-volume regional route (w_r = 0.0015) experiences an acute 200% surge (P_t/P_0 = 3.00):

- **Unweighted Naive Shift:** ΔIndex = (1/80) × (3.00 − 1.00) × 100 = **+2.50 index points**
- **DGCA Weighted Shift:** ΔAPIx = 0.0015 × (3.00 − 1.00) × 100 = **+0.30 index points**

*Takeaway: Unweighted averaging magnifies regional price anomalies by over **800%**, triggering false regulatory alerts.*

2.2 Economic Debate: Fixed vs. Adaptive Weights

In **BLS Working Paper (2021)** (Reference #8), economists evaluate fixed-weight Laspeyres/Lowe systems against adaptive/superlative formulations (e.g., Törnqvist). The findings demonstrate that while adaptive weighting captures short-term consumer substitution, computing adaptive weights on high-frequency scraped quotes introduces **severe chain drift**. APIx's fixed-weight Laspeyres/Lowe approach was chosen deliberately to maintain deterministic regulatory stability.

3. DATA CLEANING & ROBUST STATISTICAL METHODS

To prevent dynamic pricing spikes, API scraping artifacts, and seat unbundling from corrupting route price relatives, scraped fares pass through a dual-stage robust statistical sanitization pipeline:

3.1 Tukey's Interquartile Range (IQR) Inner Fences

TUKEY IQR OUTLIER BOUNDARY FORMULATION

$$IQR_{r,h,t} = Q_3(r,\ h,\ t) - Q_1(r,\ h,\ t)$$

$$\text{Acceptance Interval:}\quad \max(0,\ Q_1 - 1.5 \times IQR) \ \le\ \text{Fare}\ \le\ Q_3 + 1.5 \times IQR$$

Attribution: John W. Tukey, *Exploratory Data Analysis* (1977). Yields a 25% breakdown point against surge pricing distortions.

3.2 Median Absolute Deviation (MAD) Filter

For thin-sample corridors (*N < 10*), APIx implements the 50% breakdown-point MAD scale estimator:

HAMPEL'S MEDIAN ABSOLUTE DEVIATION (MAD)

$$MAD_{r,h,t} = \text{Median}(\ |\ p_i - \text{Median}(Fares_{r,h,t})\ |\ )$$

Normalized Scale: $\hat{\sigma}_{MAD} = 1.4826 \times MAD$ | Boundary: $|\ p_i - \text{Median}(p)\ | \leq 2.0 \times \hat{\sigma}_{MAD}$

Attribution: Frank R. Hampel, *Journal of the American Statistical Association* (1974).

4. PRECEDENTS & INDIAN REGULATORY GROUNDING

APIx is grounded in both Indian statutory governance and four decades of international price index research:

4.1 Government of India Statutory & Econometric Foundations

- **Ministry of Statistics & Programme Implementation (MoSPI) — CPI Transport Series (Reference #1):** Official national methodology under the COICOP framework for measuring transport services inflation. MoSPI utilizes digital online portal data collection and HCES household expenditure weighting. APIx acts as a high-frequency regulatory cross-check against published monthly MoSPI CPI releases.
- **DGCA — Tariff Monitoring Unit (TMU) Operational Directives (Reference #2):** Under Rule 135 of the *Aircraft Rules, 1937* and CAR Section 3, DGCA monitors airline tariff bands across 78+ domestic routes. APIx automates and scales this framework into an objective 80-route real-time index.
- **Parliamentary Standing Committee on Transport, Tourism & Culture (Report No. 328) (Reference #3):** Evaluated dynamic pricing algorithms and recommended transparent national airfare benchmarking to protect consumer welfare during peak festive surge periods.
- **Competition Commission of India (CCI) — Market Study on Civil Aviation (Reference #4):** Evaluates route-level market power (HHI), capacity discipline, and algorithmic pricing coordination risks, providing the econometric basis for the APIx Antitrust Portal (/analysts).

4.2 International Precedents

- **U.S. BLS CPI Airline Fares Factsheet (Reference #5):** Official U.S. methodology for tracking commercial airline fares in the CPI basket.
- **U.S. BLS International Price Program (IPP) (Reference #6):** Direct international precedent for passenger-volume weighted modified Laspeyres aggregation.
- **ATPI Research — Lent & Dorfman (2005) (Reference #7):** Pioneering study in the *Monthly Labor Review* validating high-frequency electronic airfare data against published CPI benchmarks.

5. LIMITATIONS & FUTURE WORK (HONEST, CITABLE)

1. **Consumer Substitution Bias (Reference #8):** Fixed-weight Laspeyres indexes do not reflect instantaneous traveler substitution away from surging routes. Future iterations will explore a **Chained Superlative Törnqvist Index** once real-time transactional APIs are linked with DGCA systems.
2. **Network Scope:** Prototype monitors 80 core domestic corridors; future expansion will incorporate regional UDAN routes to audit government Viability Gap Funding (VGF) price caps.
3. **Weight Refresh Cycles:** Regular quarterly refresh under the Lowe specification ensures seasonal alignment while preventing structural index breaks.

6. VERIFIED REFERENCE & CITATION LIST

[1] Ministry of Statistics and Programme Implementation (MoSPI), Govt. of India — *Consumer Price Index: Concepts, Definitions and Methodology for Transport Services (COICOP Framework)*, National Statistical Office (NSO).
`https://esankhyiki.mospi.gov.in/`

[2] Directorate General of Civil Aviation (DGCA), Ministry of Civil Aviation, Govt. of India — *Handbook of Civil Aviation Statistics & Tariff Monitoring Unit (TMU) Operational Directives (Rule 135, Aircraft Rules, 1937)*.
`https://www.dgca.gov.in`

[3] Parliamentary Standing Committee on Transport, Tourism & Culture, Parliament of India — *Report on Fixing of Airfares and Issues Related to Dynamic Pricing in Civil Aviation Sector (Report No. 328)*, Rajya Sabha Secretariat.
https://sansad.in

[4] Competition Commission of India (CCI), Govt. of India — *Market Study on the Civil Aviation Sector in India: Market Power, Concentration and Dynamic Pricing*.
https://www.cci.gov.in

[5] U.S. Bureau of Labor Statistics (BLS) — *Consumer Price Index: Airline Fares Factsheet*, U.S. Department of Labor.
https://www.bls.gov/cpi/factsheets/airline-fares.htm

[6] U.S. Bureau of Labor Statistics (BLS) — *Air Passenger Fares Price Indexes (IPP Methodology)*, BLS Handbook of Methods.
https://www.bls.gov/mxp/methods/air-passenger-fares.htm

[7] Janice Lent & Alan H. Dorfman (2005) — "*Air-Travel Transaction Index,*" **Monthly Labor Review**, U.S. BLS, Vol. 128, No. 6, pp. 45–54.
https://www.bls.gov/opub/mlr/2005/06/art4full.pdf

[8] U.S. Bureau of Labor Statistics (2021) — *CPI Indexes for Subsets of the Target Population: Laspeyres vs. Törnqvist Formulations and Substitution Bias*, BLS Working Paper No. 543.
https://www.bls.gov/osmr/research-papers/2021/

[9] UK Office for National Statistics (ONS) — *Consumer Price Indices Technical Manual (Methodology Appendix: Formulae used to calculate CPI and RPI)*, UK ONS Guidance.

[10] John W. Tukey (1977) — *Exploratory Data Analysis*, Addison-Wesley Series in Behavioral Science (IQR Inner Fences).

[11] Frank R. Hampel (1974) — "*The Influence Curve and its Role in Robust Estimation,*" **Journal of the American Statistical Association**, Vol. 69, No. 346, pp. 383–393 (MAD Scale Estimator).